

Laboratory Work 1 - ER Diagram

Course: Databases | Topic: Airport Reservation System

1. Entities

Airport, Airline, Flight, Passenger, Booking, BookingChangeLog, BoardingPass, Baggage, BaggageCheck, SecurityCheck.

2. Simple ER Diagram

Entity	Relationship	Entity
Airport	1 : N	Flight
Airline	1 : N	Flight
Flight	1 : N	Booking
Passenger	1 : N	Booking
Booking	1 : N	BookingChangeLog
Booking	1 : 0..1	BoardingPass
Booking	1 : 0..N	Baggage
Booking	1 : 0..N	BaggageCheck
Baggage	1 : 0..N	BaggageCheck
Passenger	1 : 0..N	BaggageCheck
Passenger	1 : N	SecurityCheck

PK = Primary Key FK = Foreign Key STR = String

3. Attributes and Data Types

Airport		
Attribute	Type	Key
airport_id	INT	PK
airport_name	STR	
country	STR	
state	STR	
city	STR	
created_at	DATETIME	
updated_at	DATETIME	

Airline		
Attribute	Type	Key
airline_id	INT	PK
airline_code	STR	
name	STR	
country	STR	
created_at	DATETIME	
updated_at	DATETIME	

Flight		
Attribute	Type	Key
flight_id	INT	PK
airline_id	INT	FK
departure_airport_id	INT	FK
arrival_airport_id	INT	FK
departing_gate	STR	
arriving_gate	STR	
scheduled_departure_time	DATETIME	
scheduled_arrival_time	DATETIME	
actual_departure_time	DATETIME	
actual_arrival_time	DATETIME	
created_at	DATETIME	
updated_at	DATETIME	

Passenger		
Attribute	Type	Key
passenger_id	INT	PK
first_name	STR	
last_name	STR	
gender	STR	
date_of_birth	DATE	
country_of_citizenship	STR	
country_of_residence	STR	
passport_number	STR	
created_at	DATETIME	
updated_at	DATETIME	

Booking		
Attribute	Type	Key
booking_id	INT	PK
flight_id	INT	FK
passenger_id	INT	FK
status	STR	
booking_platform	STR	
ticket_price	DECIMAL	
created_at	DATETIME	
updated_at	DATETIME	

BookingChangeLog		
Attribute	Type	Key
change_id	INT	PK
booking_id	INT	FK
change_type	STR	
old_value	STR	
new_value	STR	
changed_at	DATETIME	

3. Attributes and Data Types (continued)

BoardingPass		
Attribute	Type	Key
boarding_pass_id	INT	PK
booking_id	INT	FK
seat	STR	
boarding_time	DATETIME	
created_at	DATETIME	
updated_at	DATETIME	

BaggageCheck		
Attribute	Type	Key
baggage_checking_id	INT	PK
baggage_id	INT	FK
booking_id	INT	FK
passenger_id	INT	FK
check_results	STR	
created_at	DATETIME	
updated_at	DATETIME	

Baggage		
Attribute	Type	Key
baggage_id	INT	PK
booking_id	INT	FK
weight_kg	DECIMAL	
created_at	DATETIME	
updated_at	DATETIME	

SecurityCheck		
Attribute	Type	Key
security_check_id	INT	PK
passenger_id	INT	FK
check_results	STR	
created_at	DATETIME	
updated_at	DATETIME	

4. Relationships and Cardinality

From	To	Cardinality	Meaning
Airport	Flight	1 : N	departure airport
Airport	Flight	1 : N	arrival airport
Airline	Flight	1 : N	operates
Flight	Booking	1 : N	has
Passenger	Booking	1 : N	makes
Booking	BookingChangeLog	1 : N	change history
Booking	BoardingPass	1 : 0..1	issues
Booking	Baggage	1 : 0..N	has
Booking	BaggageCheck	1 : 0..N	has
Baggage	BaggageCheck	1 : 0..N	checked in
Passenger	BaggageCheck	1 : 0..N	owns
Passenger	SecurityCheck	1 : N	passes

5. Notation

Symbol	Meaning
PK	Primary key
FK	Foreign key
1	Exactly one
N	Many
0..1	Zero or one
0..N	Zero or many

Airport Reservation System — ER Diagram

This ER diagram represents an airport reservation system. It shows the main entities used to manage airports, airlines, flights, passengers, bookings, boarding passes, baggage, baggage checks, security checks, and booking changes. The relationships explain how these entities are connected: airlines operate flights, passengers make bookings, bookings may have baggage and boarding passes, and security and baggage checks are recorded for airport processing.

